

Guillermo Fernando Farfán Romero

Senior Database Administrator (DBA) & Data Engineer

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Portfolio: [gfarfan.dev](#) | [LinkedIn](#) | [GitHub](#) | [Credly](#)

PROFESSIONAL SUMMARY

Senior Database Administrator (DBA) and Data Engineer with 8+ years of experience architecting, administering, optimizing, and scaling relational, NoSQL, vector, and high-throughput Data Warehouse platforms. Specialist in **Microsoft SQL Server**, **PostgreSQL (with pgvector for semantic search/AI)**, advanced **Performance Tuning** (execution plan analysis, table partitioning, deep indexing, deadlock resolution), cloud data architectures on **Google Cloud Platform (Cloud SQL, BigQuery)**, and automated ELT/ETL pipelines with **Apache Airflow, dbt, and Python (Polars, PyArrow, SQLAlchemy)**.

Combines deep technical expertise in data persistence, governance, and business continuity (DRP/BCP with strict RPO/RTO) with strategic team leadership, backed by a dual background in Information Systems and Labor Relations.

TECHNICAL SKILLS & COMPETENCIES

- Database Engines (RDBMS & Vector):** Microsoft SQL Server (2012-2022, AlwaysOn Availability Groups, SSIS, SSRS, T-SQL, CDC), PostgreSQL (12-16, pgvector, Declarative Partitioning, Streaming Replication, PgBouncer, Connection Pooling), MySQL, MariaDB.
- Data Warehousing, Lakehouse & NoSQL:** Google BigQuery (Partitioning, Clustering, Slot Optimization, Star Schema / Kimball, BigQuery Vector Search), Redis (Distributed Caching, Clustering, Pub-Sub, In-Memory Storage), Google Cloud Pub/Sub, MongoDB, Cloud Firestore.
- Data Engineering & Orchestration:** Apache Airflow, Python (Polars, PyArrow, Pandas, SQLAlchemy 2.0, PyODBC), Alembic (Zero-Downtime Migrations), dbt (Data Modeling & Testing), Data Contracts, OpenLineage, Data Quality, Batch & Streaming ETL/ELT Pipelines, Celery, Bash Scripting.
- Cloud Infrastructure & Storage:** Google Cloud Platform (Cloud SQL, BigQuery, Compute Engine Persistent Disk, Cloud Storage), Linux (SLES 15 SP7, Ubuntu), XFS File System, LVM, IOPS & Throughput Tuning.
- Diagnostics & Performance Tuning:** Query Execution Plans, SQL Profiler, Extended Events, Index Fragmentation Management, Slow Query Logs, Deadlock/Blocking Resolution, Real-time Observability with IBM Instana APM and OpenTelemetry.
- Security & Business Continuity:** Transactional Replication, Automatic Failover, Strict RPO/RTO Backup/Restore Strategies, TDE/SSL Encryption, Role-Based Access Control (RBAC), Data Governance.

PROFESSIONAL EXPERIENCE

Technical Lead & Senior Systems Engineer (Data Architecture Focus) | Reforest Latam

May 2026 – Present | Full-time (Remote)

- Architected and implemented a distributed caching and in-memory persistence layer with Redis** for environmental telemetry platforms, driving a **45% speedup in concurrent analytical and transactional queries**.
- Standardized 100% of database lifecycle and schema version control**, establishing automated schema migrations with **Alembic** and CI/CD pipelines for **zero-downtime deployments** without service interruptions.
- Led relational persistence design and governance in PostgreSQL** for high-concurrency platforms, enforcing referential integrity, declarative table partitioning, and strategic normalization.
- Directed and mentored an engineering team of 6**, defining SQL query optimization guidelines, complex transaction tuning, and schema decoupling standards.

Senior Data Infrastructure Consultant & DBA | Independent Consulting & Enterprise Projects

January 2021 – Present | Strategic B2B Technology Consultant (Independent / Contract / Remote)

- Engineered and deployed automated ELT pipelines with Apache Airflow and Python (Polars, PyArrow, SQLAlchemy)**, cutting **batch processing windows by 60%** across **+500 GB daily** transactional datasets loaded into **Google BigQuery** (optimized via partitioning, clustering, and Star Schema modeling).
- Maximized storage I/O throughput and reduced latency** across Linux servers (SLES 15 SP7 / Ubuntu) and Cloud SQL for enterprise retail and mining clients through block-aligned XFS and advanced LVM management.
- Implemented real-time transactional observability with IBM Instana APM and OpenTelemetry**, tracking TPS, p95/p99 query latencies, and eliminating connection pool bottlenecks.
- Enforced database security hardening and regulatory compliance**, implementing RBAC policies, data-at-rest and in-transit encryption (TDE/SSL), and production security audits.

Senior Software Engineer & Database Administrator | University of Buenos Aires (UBA)

March 2018 – February 2024 | Institutional Systems Specialist (Part-time / Buenos Aires, Argentina)

- Spearheaded comprehensive performance tuning across 8+ TB of mission-critical data** (+50,000 active users), restructuring indexes, optimizing execution plans, and resolving deadlocks to **cut p95 query latency by 55%**.
- Guaranteed RPO < 15 min and RTO < 1 hour** through High Availability (HA) replication architectures and robust Disaster Recovery Plans (DRP/BCP).
- Developed automated Python ETL pipelines**, accelerating data cleansing, validation, and migration from legacy databases to modern relational schemas.

Systems Infrastructure & Operations Coordinator | Government of the City of Buenos Aires (GCBA)

January 2020 – December 2023 | Full-time (Buenos Aires, Argentina)

- **Optimized maintenance schedules, native compression, and differential backup rotation on SQL Server and PostgreSQL**, achieving a **35% reduction in overnight backup windows**.
- **Maintained 99.9% operational database uptime (SLA)** across municipal citizen portals serving 100,000+ monthly users.
- **Implemented data access governance and auditing**, establishing historical data retention policies and strict information security compliance.

EDUCATION

- **Associate Degree in Information Systems** | National Technological University (UTN) (*In Progress — Expected Graduation: December 2026*)
- **Bachelor's Degree in Labor Relations & Human Resources** | University of Buenos Aires (UBA) (*Graduated — Focus on Organizational Leadership, Talent Management & High-Performance Team Dynamics*)

KEY CERTIFICATIONS (100% VERIFIABLE ON CREDLY)

- **Professional Data & Backend Certifications (Credly):** Google Data Analytics Professional Certificate | Meta Back-End Developer Certificate | IBM DevOps and Software Engineering Professional Certificate.
- **Cloud & Persistence Specializations:** Google Cloud Fundamentals: Core Infrastructure (Google Cloud) | IBM Full Stack Software Developer Certificate | Oracle Next Education F2 Back-End.

LANGUAGES

- **Spanish:** Native
- **English:** Full Professional Proficiency (C1 / Full technical fluency for data architecture, live engineering syncs, RFCs, and technical documentation)